

Searching for the digital dividend: A debiased machine-learning sieve estimator for heterogeneous effects of digital adoption in panel data

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Abstract

Whether digital adoption raises productivity is usually assessed with two-way fixed-effects regressions that impose a single, linear effect and a linear control function. Both restrictions are at odds with theories in which the returns to digital technologies depend on absorptive capacity and distance to the technological frontier. This paper develops Panel-DOSE, a debiased orthogonal sieve estimator for panel data that recovers the whole function linking the effect of a continuous digital treatment to a moderating variable. The estimator combines Mundlak-augmented machine-learning nuisance functions cross-fitted by country, a two-way within transformation of the orthogonalised residuals, and a penalised B-spline sieve whose smoothness is chosen by generalised cross-validation. It delivers pointwise and simultaneous confidence bands, group-average effects and a one-degree-of-freedom test of heterogeneity. We show that the estimating equation is Neyman-orthogonal and exactly invariant to additive unit and time errors in the nuisance functions, and that, when the moderator is a control variable, only two nuisance functions are needed whatever the sieve dimension. In Monte Carlo experiments with nonlinear confounding, Panel-DOSE attains the lowest integrated error in every non-constant design, reducing it by 60–80% relative to linear fixed-effects models and boosted R-learners. Applied to 141 countries over 1996–2025, the estimator finds no aggregate digital dividend: the average effect of internet adoption

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on subsequent labour-productivity growth is small, negative and insignificant, which rules out gains larger than about 0.13 percentage points per ten points of adoption, and it does not vary with schooling. Heterogeneity along the development dimension is not robust: its sign differs between World Bank data and the 10.0 vintage of the Penn World Table, and it vanishes in the 11.0 vintage for the same years, whereas a linear fixed-effects interaction reports it with spurious precision. Claims about where the digital dividend arrives should therefore be treated with caution.

Keywords: Digital adoption, Labor productivity, Double machine learning, Heterogeneous treatment effects, Panel data, Varying-coefficient model

JEL: C14, C23, C55, O33, O47

1. Introduction

The expectation that digital technologies deliver a “digital dividend” in the form of faster productivity growth is central to development strategies and to much of the digital-economy literature (World Bank, 2016; Goldfarb and Tucker, 2019). The empirical record is more mixed than this expectation suggests. Firm- and industry-level studies document large returns to information technology, but mainly where it is combined with organisational change and skilled labour (Brynjolfsson and Hitt, 2000; Bresnahan et al., 2002). Cross-country evidence ranges from sizeable growth effects of broadband diffusion in advanced economies (Czernich et al., 2011) to weak or insignificant effects in developing and emerging economies (Niebel, 2018; Vu et al., 2020). Recent contributions to this journal likewise emphasise that the economic consequences of digitalisation depend on structural and institutional conditions (Banga et al., 2026; Chen and Li, 2026).

A common thread in these findings is heterogeneity. Classic growth theory predicts that the payoff from a new technology depends on the capacity to absorb it, whether this is captured by human capital (Nelson and Phelps, 1966; Benhabib and Spiegel, 1994; Cohen and Levinthal, 1990) or by distance to the technological frontier (Acemoglu et al., 2006; Comin and Hobijn, 2010). The productivity J-curve of Brynjolfsson et al. (2021) adds that general-

purpose technologies such as the internet require complementary intangible investment that depresses measured productivity before raising it. Taken together, these arguments imply that the effect of digital adoption is a *function* of the adopting economy’s characteristics, possibly non-monotone and possibly negative over part of its domain; arguments about “leapfrogging” point in the opposite direction, predicting the largest gains where information and transaction costs are initially highest.

The standard empirical toolkit is poorly suited to estimating such a function. Two-way fixed-effects (TWFE) regressions deliver one number and, when heterogeneity is modelled at all, do so through a linear interaction term. They also specify the control function linearly, although the determinants of both productivity growth and digital adoption—income, capital deepening, demography, openness—interact nonlinearly. When the control function is misspecified and correlated with adoption, the resulting bias contaminates the average effect and, more seriously, its apparent heterogeneity. Double/debiased machine learning (DML; Chernozhukov et al., 2018) addresses the control-function problem, and its panel extensions (Clarke and Polsell, 2023) deal with fixed effects, but they target a constant parameter. Methods for heterogeneous effects, such as causal forests and the R-learner (Wager and Athey, 2018; Athey et al., 2019; Nie and Wager, 2021), have been developed for cross-sectional data with binary treatments; they do not exploit the within-country variation that identifies effects in macro panels and they provide limited inference on the shape of the effect function.

This paper proposes an estimator that fills this gap. Panel-DOSE (*Panel Debiased Orthogonal Sieve Estimator*) targets the varying-coefficient model

$$Y_{it} = \theta(Z_{it}) D_{it} + g(X_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (1)$$

in which D_{it} is a continuous digital-adoption measure, Z_{it} is a moderator such as human capital or initial productivity, $g(\cdot)$ is an unknown control function and (α_i, λ_t) are country

and year effects. The effect function $\theta(\cdot)$ is approximated by a penalised cubic B-spline sieve (Eilers and Marx, 1996), and the sieve coefficients are estimated from a Neyman-orthogonal moment built on machine-learning residuals. The estimator has four ingredients. First, the nuisance learners are fed Mundlak (time-average) features, which absorb the part of the fixed effect that is correlated with observables (Mundlak, 1978; Wooldridge, 2010). Second, the nuisance functions are cross-fitted with folds formed by countries, so that no country’s outcome is used to predict itself. Third, the cross-fitted residuals are two-way demeaned, which makes the moment exactly invariant to any additive country or year error in the machine-learning fits. Fourth, the smoothness of $\theta(\cdot)$ is chosen by generalised cross-validation, and inference relies on a cluster-robust sandwich, simultaneous (sup- t) bands (Montiel Olea and Plagborg-Møller, 2019), group-average effects and a best-linear-projection test of heterogeneity (Semenova and Chernozhukov, 2021).

The paper makes three contributions. The first is methodological. We extend orthogonal sieve estimation of causal functions (Semenova and Chernozhukov, 2021) to fixed-effects panels with a continuous treatment, prove that the resulting moment is Neyman-orthogonal and invariant to additive nuisance errors, and establish asymptotic normality under conditions that are standard for DML with clustered data. We also show a computational property that is specific to the moderator being a control variable: all orthogonalised regressors can be generated from two nuisance functions, $\mathbb{E}[Y \mid \cdot]$ and $\mathbb{E}[D \mid \cdot]$, so the cost of the estimator does not grow with the sieve dimension and alternative sieves can be compared at no additional cost.

The second contribution is evidence on finite-sample performance. In a Monte Carlo study calibrated to a macro panel of 100 countries observed for 24 years, with fixed-effect and nonlinear confounding, Panel-DOSE attains the lowest integrated root-mean-squared error in all designs with a non-constant effect. It improves markedly on TWFE with linear interactions, on constant-effect panel DML, on a boosted R-learner applied to the same orthogonalised residuals, and on a pooled DML sieve that ignores the panel structure. The

ablation shows that the fixed-effects treatment of the nuisance step, rather than machine learning per se, is responsible for most of the gain.

The third contribution is substantive. We apply the estimator to 141 economies over 1996–2025, combining World Bank productivity and ITU adoption data with UNDP schooling data, and compare the results with two vintages of the Penn World Table (Feenstra et al., 2015): version 11.0 for 1996–2023 and version 10.0 for 1996–2019. Conditional on flexible controls, the average effect of lagged internet adoption on labour-productivity growth is small, negative and statistically insignificant, and the confidence interval excludes the large effects often invoked in policy debates. The effect does not vary with schooling. Along the development dimension, the baseline estimates suggest larger effects in the least productive economies, but the gradient is not robust: PWT 10.0 data point to the opposite, a strongly negative effect in the least productive economies, and in PWT 11.0 for the same years the gradient is small and insignificant. The revision of the productivity data alone thus overturns a result that is significant at the 1% level in the earlier vintage. A linear TWFE interaction, by contrast, reports a highly significant gradient, illustrating how a restrictive control function can manufacture precision. The results suggest that the digital dividend is not automatic and that evidence on where it arrives is more fragile than conventional estimates imply.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 presents the model, the Panel-DOSE algorithm and its properties. Section 4 reports the Monte Carlo evidence. Section 5 presents the empirical application, Section 6 discusses implications and limitations, and Section 7 concludes. Proofs are collected in Appendix A and data and implementation details in Appendix B.

2. Related literature

2.1. Digital adoption, absorptive capacity and productivity

Growth accounting attributes a large share of the US productivity resurgence of the late 1990s to the production and use of information technology (Jorgenson et al., 2008). Outside the frontier, evidence on the productivity effects of digital technologies is less conclusive. Czernich et al. (2011) use an instrumental-variable approach based on pre-existing telecommunication networks and find that a ten-percentage-point increase in broadband penetration raised annual per-capita growth by 0.9–1.5 percentage points in OECD countries. Niebel (2018) finds that the output elasticity of ICT capital does not differ significantly between developing, emerging and developed economies, and that emerging and developing economies do not gain more from ICT investment than developed ones. Vu et al. (2020) survey more than a hundred studies and conclude that the growth effect of ICT is positive on average but heterogeneous across countries, periods and technologies. Micro evidence points in the same direction: Hjort and Poulsen (2019) show that the arrival of fast internet in Africa increased employment, particularly in skilled occupations.

Several theories predict this heterogeneity. In Nelson and Phelps (1966) and Benhabib and Spiegel (1994), human capital governs the speed at which economies adopt and exploit new technologies; in the organisational literature, absorptive capacity plays the same role at the firm level (Cohen and Levinthal, 1990). In Acemoglu et al. (2006), the optimal growth strategy changes with distance to the frontier, so the same technology can have different returns at different stages of development. Comin and Hobijn (2010) document that adoption lags vary widely across countries and technologies, implying that the intensity of use, not just availability, determines the gains. Brynjolfsson et al. (2021) formalise the J-curve: because general-purpose technologies need complementary intangible investment that is poorly measured, measured productivity growth first falls and then rises as the technology spreads. Measurement concerns have also been raised as an explanation for the post-2005 productivity slowdown, although Syverson (2017) and Byrne et al. (2016) find

that mismeasurement accounts for only a small part of it.

Empirically, heterogeneity is usually handled through sample splits by income group (Niebel, 2018), linear interaction terms, or threshold regressions (Hansen, 1999). Sample splits are coarse and interactions impose linearity; threshold models impose a step function and are typically combined with linear controls. None of these approaches accommodates a flexible control function, and inference on the shape of the effect is usually limited to the coefficients of the chosen parametric form. Recent machine-learning contributions to this journal, for example on corporate carbon emissions (Xu et al., 2026), use flexible learners for prediction rather than for estimating causal or structural effects.

2.2. Machine learning for causal and structural parameters

Machine learning has become part of the applied econometrician’s toolkit (Mullainathan and Spiess, 2017; Athey and Imbens, 2019). DML (Chernozhukov et al., 2018) generalises the partialling-out estimator of Robinson (1988): it estimates the nuisance functions with any sufficiently accurate learner, uses cross-fitting to avoid overfitting bias and bases inference on a Neyman-orthogonal score that is insensitive to first-order errors in the nuisance estimates. Clarke and Polsell (2023) adapt DML to static panels with fixed effects and compare correlated-random-effects, within-group and first-difference implementations. Their target is a constant coefficient.

Heterogeneous effects have been studied mainly in cross-sectional settings with binary treatments. Causal forests (Wager and Athey, 2018; Athey et al., 2019) adapt random forests to estimate conditional average treatment effects; the R-learner of Nie and Wager (2021) minimises a residual-on-residual loss with a flexible function class. Semenova and Chernozhukov (2021) show that projecting an orthogonal signal onto a growing series basis yields asymptotically normal estimates of causal functions and their best linear predictors. Varying-coefficient models (Hastie and Tibshirani, 1993; Cai et al., 2000) are the classical semiparametric counterpart, estimated by kernel or spline methods under parametric control functions. Series estimation theory (Chen, 2007; Belloni et al., 2015) supplies the pointwise

and uniform asymptotics for such sieves.

Panel-DOSE sits at the intersection of these strands. Relative to Clarke and Polsell (2023), it targets a function rather than a scalar and combines the correlated-random-effects and within-group strategies instead of choosing between them. Relative to Semenova and Chernozhukov (2021) and Nie and Wager (2021), it handles two-way fixed effects, cluster dependence and a continuous treatment, and exploits a structure-aware residualisation that keeps the cost independent of the sieve dimension. Relative to varying-coefficient and threshold models, it does not require the control function to be linear.

3. Methodology

3.1. Model and target parameters

We observe a panel $\{(Y_{it}, D_{it}, X_{it}) : i = 1, \dots, G; t = 1, \dots, T_i\}$ of G countries, where Y_{it} is an outcome (productivity growth), $D_{it} \in \mathbb{R}$ is a continuous digital-adoption measure, and $X_{it} \in \mathbb{R}^p$ collects controls that contains the moderator Z_{it} . We work with model (1). The function $\theta(z)$ is the marginal effect of adoption on the outcome for an economy with moderator value z : a one-unit increase in D changes Y by $\theta(z)$, holding the controls and fixed effects constant. We also consider a dose-response variant,

$$Y_{it} = f(D_{it}) + g(X_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (2)$$

whose target is the marginal effect $f'(d)$, which tells whether the returns to adoption diminish or rise with the level of adoption itself.

Three summaries of $\theta(\cdot)$ are useful in practice. The *average effect* is $\bar{\theta} = \mathbb{E}[\theta(Z_{it})]$. The *group-average effect* for a set A of moderator values is $\bar{\theta}_A = \mathbb{E}[\theta(Z_{it}) \mid Z_{it} \in A]$; we use the terciles of Z . The *best linear projection* (BLP) slope γ solves $\min_{a, \gamma} \mathbb{E}[(\theta(Z_{it}) - a - \gamma Z_{it})^2]$ and gives a one-degree-of-freedom summary of how the effect changes with the moderator (Semenova and Chernozhukov, 2021).

3.2. Identification

Write $F_{it} = (X_{it}, \bar{X}_i, \bar{D}_i, t)$ for the controls augmented by their country averages $\bar{X}_i = T_i^{-1} \sum_t X_{it}$ and $\bar{D}_i = T_i^{-1} \sum_t D_{it}$, and a time index.

Assumption 1 (Conditional strict exogeneity). $\mathbb{E}[\varepsilon_{it} \mid D_{i1}, \dots, D_{iT_i}, X_{i1}, \dots, X_{iT_i}, \alpha_i] = 0$ for all t .

Assumption 2 (Correlated random effects in the nuisance step). $\mathbb{E}[\alpha_i \mid F_{it}] = h(\bar{X}_i, \bar{D}_i)$ for some function h that does not depend on t .

Assumption 3 (Nuisance sufficiency). $\mathbb{E}[\Phi_{it} \mid F_{i1}, \dots, F_{iT_i}] = \mathbb{E}[\Phi_{it} \mid F_{it}]$ for all t .

Assumption 1 is the usual strict-exogeneity condition of static fixed-effects models; in the application it motivates entering digital adoption with a lag and it is discussed further in Section 6. Assumption 2 is a nonparametric version of the Mundlak device: the conditional mean of the country effect given the augmented controls may be any function of the country averages (Mundlak, 1978; Wooldridge, 2010). It is weaker than the linear Mundlak projection, and the part of α_i that it does not explain, $a_i = \alpha_i - h(\bar{X}_i, \bar{D}_i)$, is removed by the within transformation described below. Assumption 3 requires that the augmented controls of period t summarise the information in the whole control path that is relevant for predicting adoption; the Mundlak averages in F_{it} are included precisely to carry this country-level information. Finally, D_{it} must retain variation after conditioning on F_{it} and removing country and year means; this overlap condition is formalised as the invertibility of the matrix J in Assumption 4.

Let $b(z) = (b_1(z), \dots, b_K(z))'$ be a cubic B-spline basis with interior knots at quantiles of Z and write $\theta(z) = b(z)'\beta + r_K(z)$ with approximation error r_K . Define the K “sieve regressors” $\Phi_{it} = D_{it} b(Z_{it})$ and the nuisance functions $\ell(F) = \mathbb{E}[Y \mid F]$ and $m(F) = \mathbb{E}[\Phi \mid F]$. Let $\mathcal{M}$ be the two-way within operator that removes country and year means from a

variable (for unbalanced panels, by alternating projections). The population moment is

$$\psi(\beta; \ell, m) = \mathbb{E} \left[\sum_t \mathcal{M}(\Phi - m(F))_{it} \left\{ \mathcal{M}(Y - \ell(F))_{it} - \mathcal{M}(\Phi - m(F))'_{it} \beta \right\} \right] = 0. \quad (3)$$

Under Assumptions 1–3 and $r_K = 0$, $Y - \ell(F) - (\Phi - m(F))' \beta = a_i + \varepsilon_{it}$, because λ_t is a function of the time index in F and is absorbed by ℓ . The term a_i is removed by $\mathcal{M}$ and ε_{it} is mean-independent of the whole path of F , so the true β solves (3).

3.3. Two structural properties

Proposition 1 (Orthogonality and fixed-effect invariance). *Let ψ be defined by (3) and let Assumptions 1–3 hold. (i) Neyman orthogonality: for any directions (δ_ℓ, δ_m) in the nuisance space, $\partial_r \psi(\beta_0; \ell_0 + r\delta_\ell, m_0 + r\delta_m)|_{r=0} = 0$. (ii) Exact invariance: for any country-specific and year-specific constants (c_i, c_t) and $(d_i, d_t) \in \mathbb{R}^K$, $\psi(\beta; \ell + c_i + c_t, m + d_i + d_t) = \psi(\beta; \ell, m)$ for all β .*

Part (i) is the property that makes DML insensitive to first-order errors in the machine-learning fits. Part (ii) is specific to the panel construction: any error of the learners that takes the form of a country or year shift—a natural failure mode when trees approximate a fixed effect through country averages—has no effect on the estimate at all, so the product-rate condition of DML only needs to hold for the within-country, within-year component of the nuisance error. Section 4 shows that this matters in finite samples.

Proposition 2 (Structure-aware residualisation). *If Z_{it} is an element of F_{it} , then $m_k(F) = b_k(Z) \mu(F)$ for $k = 1, \dots, K$, where $\mu(F) = \mathbb{E}[D \mid F]$. Hence the orthogonalised sieve regressors are $b(Z_{it})\{D_{it} - \mu(F_{it})\}$, and the estimator requires only the two nuisance functions ℓ and μ , whatever the sieve dimension K .*

Proposition 2 follows from $\mathbb{E}[D b_k(Z) \mid F] = b_k(Z) \mathbb{E}[D \mid F]$ when Z is measurable with respect to F . It reduces the number of learners from $K + 1$ to two, avoids the instability of fitting K separate regressions for highly collinear spline regressors, and allows the sieve

Algorithm 1 Panel-DOSE

Require: Panel $(Y_{it}, D_{it}, X_{it}, Z_{it})$; learner $\mathcal{L}$; folds L ; repetitions S ; knots q

- 1: Build $F_{it} = (X_{it}, \bar{X}_i, \bar{D}_i, \bar{Z}_i, t)$ and the B-spline basis $b(\cdot)$ with q interior knots at quantiles of Z
 - 2: **for** $s = 1, \dots, S$ **do**
 - 3: Randomly partition the *countries* into L folds $I_1, \dots, I_L$
 - 4: **for** $l = 1, \dots, L$ **do**
 - 5: Fit $\hat{\ell}^{(-l)}$ and $\hat{\mu}^{(-l)}$ by $\mathcal{L}$ on countries not in I_l
 - 6: For $i \in I_l$: $\hat{e}_{it}^Y = Y_{it} - \hat{\ell}^{(-l)}(F_{it})$, $\hat{e}_{it}^\Phi = b(Z_{it})\{D_{it} - \hat{\mu}^{(-l)}(F_{it})\}$
 - 7: **end for**
 - 8: Two-way demean: $\tilde{Y} = \mathcal{M}\hat{e}^Y$, $\tilde{\Phi} = \mathcal{M}\hat{e}^\Phi$
 - 9: Choose $\hat{\lambda}_s$ by GCV and compute $\hat{\beta}_s$ from (4)
 - 10: Compute the cluster-robust covariance $\hat{V}_s$ from (5)
 - 11: **end for**
 - 12: $\hat{\beta} = \text{median}_s \hat{\beta}_s$; $\hat{V} = \text{median}_s \{\hat{V}_s + (\hat{\beta}_s - \hat{\beta})(\hat{\beta}_s - \hat{\beta})'\}$
 - 13: **return** $\hat{\theta}(z) = b(z)'\hat{\beta}$, bands, group averages and BLP slope
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(number of knots, penalty, or a linear basis for the BLP) to be changed after the nuisance step without refitting. In the dose-response model (2), Φ is a basis in D and the K conditional means are learned separately.

3.4. The Panel-DOSE algorithm

Algorithm 1 summarises the estimator. The final stage minimises a penalised least-squares criterion in the orthogonalised variables,

$$\hat{\beta}_\lambda = \arg \min_{\beta} \sum_{i,t} (\tilde{Y}_{it} - \tilde{\Phi}'_{it}\beta)^2 + \lambda n \beta' P \beta, \quad P = \Delta'_2 \Delta_2, \quad (4)$$

where $\tilde{Y} = \mathcal{M}(Y - \hat{\ell})$, $\tilde{\Phi} = \mathcal{M}(\Phi - \hat{m})$, $n = \sum_i T_i$, and Δ_2 is the second-difference matrix of the P-spline approach of Eilers and Marx (1996). Because the null space of P contains the coefficients of linear functions, a large λ shrinks $\theta(\cdot)$ towards a linear function rather than towards zero. The penalty is chosen by generalised cross-validation (Craven and Wahba, 1979), $\text{GCV}(\lambda) = n \text{RSS}(\lambda) / \{n - \text{tr}(H_\lambda)\}^2$, with $H_\lambda = \tilde{\Phi}(\tilde{\Phi}'\tilde{\Phi} + \lambda n P)^{-1}\tilde{\Phi}'$.

Cross-fitting by countries rather than by observations is essential in a panel. With observation-level folds, the learner is trained on other years of the same country, so the

fitted value of Y_{it} partly reproduces its own country effect and the residuals are too small; country-level folds keep the training and evaluation samples independent under cluster sampling. Repeating the split S times and taking the median, with the variance adjustment of Chernozhukov et al. (2018), removes the dependence of the estimate on a particular partition.

3.5. Inference

Let $A_\lambda = \tilde{\Phi}'\tilde{\Phi} + \lambda nP$ and $\hat{u}_{it} = \tilde{Y}_{it} - \tilde{\Phi}'_{it}\hat{\beta}$. The cluster-robust covariance estimator is

$$\hat{V} = c_n A_\lambda^{-1} \left(\sum_{i=1}^G \hat{s}_i \hat{s}_i' \right) A_\lambda^{-1}, \quad \hat{s}_i = \sum_t \tilde{\Phi}_{it}' \hat{u}_{it}, \quad (5)$$

with the small-sample factor $c_n = \frac{G}{G-1} \frac{n-1}{n-K}$ (Cameron and Miller, 2015). For any linear functional $\omega(\beta) = w'\beta$ —the value $\hat{\theta}(z) = b(z)'\hat{\beta}$, the average effect with $w = n^{-1} \sum_{it} b(Z_{it})$, a group average, or a difference of group averages—the standard error is $(w'\hat{V}w)^{1/2}$. Simultaneous bands over a grid $z_1, \dots, z_J$ use the critical value c_{1-a} equal to the $(1-a)$ quantile of $\max_j |\xi_j|$, where $\xi \sim N(0, \hat{\Sigma})$ and $\hat{\Sigma}$ is the correlation matrix of $\{\hat{\theta}(z_j)\}$ (Montiel Olea and Plagborg-Møller, 2019). The BLP slope is computed by re-solving the final stage with a linear basis $b(z) = (1, z)'$, which by Proposition 2 requires no new nuisance fits.

Assumption 4 (Regularity). (a) Countries are independent, $G \rightarrow \infty$, and $T_i \leq \bar{T} < \infty$. (b) The cross-fitted nuisance estimators satisfy $\|\hat{\mu} - \mu\|_2 (\|\hat{\mu} - \mu\|_2 + \|\hat{\ell} - \ell\|_2) = o_p(G^{-1/2})$ and $\|\hat{\ell} - \ell\|_2 = o_p(1)$, where by Proposition 1(ii) the norms need only apply to the within-transformed errors. (c) $J = \mathbb{E}[\sum_t \tilde{\Phi}_{it}^0 \tilde{\Phi}_{it}^{0'}]$ is positive definite, where $\tilde{\Phi}^0$ uses the true nuisance functions, and fourth moments are bounded. (d) K is fixed and $\lambda_G \sqrt{G} \rightarrow 0$.

Proposition 3 (Asymptotic normality). Under Assumptions 1–4 and the conditions of Proposition 2, $\sqrt{G}(\hat{\beta} - \beta_K) \xrightarrow{d} N(0, J^{-1}\Omega J^{-1})$, where β_K is the sieve pseudo-true value and $\Omega = \mathbb{E}[s_i s_i']$ with $s_i = \sum_t \tilde{\Phi}_{it}^0 u_{it}$, and $G\hat{V} \xrightarrow{p} J^{-1}\Omega J^{-1}$.

Remark 1. Proposition 3 treats the sieve dimension as fixed, so $\hat{\theta}(z)$ is centred at the best sieve approximation $b(z)'\beta_K$. Inference on $\theta(z)$ itself requires $K \rightarrow \infty$ with the approximation

bias negligible relative to the standard error; the conditions of Belloni et al. (2015) and Semenova and Chernozhukov (2021) for series projections of orthogonal signals then deliver pointwise and uniform validity. The average effect, group averages and the BLP slope are smooth functionals that do not require undersmoothing. In practice, the penalty selected by GCV trades a small smoothing bias for a large variance reduction; Section 4 quantifies the effect of this choice on coverage.

4. Monte Carlo evidence

4.1. Design

The simulation mimics a macro panel of $G = 100$ countries observed for $T = 24$ years. Each country has a fixed effect $\alpha_i \sim N(0, 1)$ and there is a random-walk year effect. Five controls follow $X_{it} = 0.6X_{i,t-1} + 0.4(\eta_{it} + 0.5\alpha_i)$ with $\eta_{it} \sim N(0, I_5)$, so they are persistent and correlated with the fixed effect. The moderator $Z_{it} \in [0, 1]$ trends slowly around a country-specific level. Digital adoption is a logistic diffusion index,

$$D_{it} = \Lambda(-3 + 0.22t + 0.6\alpha_i + 0.8\sin(1.5X_{1it}) + 0.6X_{2it}\mathbb{I}\{X_{3it} > 0\} + 1.5Z_{it} + v_{it}),$$

where Λ is the logistic function and $v_{it} \sim N(0, 1)$, and the outcome is $Y_{it} = \theta(Z_{it})D_{it} + g(X_{it}, Z_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}$ with $\varepsilon_{it} \sim N(0, 0.25)$ and the nonlinear control function

$$g = \rho\{1.5\sin(1.5X_{1it}) + X_{2it}\mathbb{I}\{X_{3it} > 0\} + 0.5|X_{4it}| + 3.2(Z_{it} - 0.5)^2\}.$$

Adoption and the outcome therefore share country effects, a time trend and nonlinear functions of the controls. We consider four effect functions: *constant* $\theta(z) = 1$; *linear* $\theta(z) = -0.5 + 3z$; *threshold* $\theta(z) = -0.5 + 2.5/(1 + e^{-12(z-0.5)})$, a smooth J-curve-type transition; and *hump* $\theta(z) = 2 - 8(z - 0.5)^2$. The baseline sets $\rho = 1$; sensitivity analyses vary the confounding strength ($\rho \in \{0.5, 2\}$) and the number of countries ($G \in \{50, 200\}$) for the threshold design.

We compare seven estimators: (1) *TWFE*, a linear two-way fixed-effects regression with a constant effect; (2) *TWFE-interaction*, which adds $D \times Z$ and $D \times Z^2$; (3) *FE-DML (constant)*, the panel DML estimator with Mundlak features, country-level cross-fitting and within transformation, which is Panel-DOSE with a single basis function, in the spirit of Clarke and Polselli (2023); (4) *FE-DML + R-learner*, which fits $\theta(\cdot)$ to the same orthogonalised residuals with gradient boosting and the R-learner loss (Nie and Wager, 2021); (5) *Pooled DML-sieve*, the sieve estimator of Semenova and Chernozhukov (2021) applied to the pooled data without Mundlak features or within transformation; (6) *Panel-DOSE (unpenalised)*; and (7) *Panel-DOSE*. All machine-learning estimators use the same LightGBM learner (Ke et al., 2017) with 600 trees, five country-level folds and, for the sieve estimators, cubic B-splines with four interior knots and two repetitions of the sample split. Performance is measured on a grid of 17 points in $[0.1, 0.9]$ by the integrated root-mean-squared error $\text{IRMSE} = \{J^{-1} \sum_j (\hat{\theta}(z_j) - \theta(z_j))^2\}^{1/2}$ and the integrated bias, by the bias and RMSE of the average effect, and by the coverage of nominal 95% pointwise and simultaneous bands. Each design is replicated 100 times (50 in the sensitivity analyses).

4.2. Results

Table 1 and Figure 1 report the results for the baseline designs. Four findings stand out.

First, the linear fixed-effects estimators are severely biased. Because the control function is nonlinear and correlated with adoption, TWFE and TWFE-interaction overstate the effect by 0.5–0.9 units on average, and their confidence intervals almost never cover the truth. The interaction model does not help even when the true $\theta(\cdot)$ is exactly quadratic (the constant, linear and hump designs, in which its errors are identical because the effect function lies in the span of its regressors): its problem is the control function, not the flexibility of the effect.

Second, handling the panel structure in the nuisance step is essential. The pooled DML sieve uses the same learner and the same sieve as Panel-DOSE but neither Mundlak features nor the within transformation. Its integrated bias is about 0.6 in every design, of the same order as that of TWFE, because the learners cannot separate the country effects from the

controls and the orthogonalised treatment still carries the correlation between adoption and α_i . Once the Mundlak features and the within step are added, the integrated bias falls to 0.05–0.07. The remaining bias is common to all panel-DML estimators and reflects the finite-sample regularisation bias of the nuisance learners; it is small relative to the effect.

Third, among the estimators that handle fixed effects, the sieve dominates the alternatives whenever the effect is heterogeneous. Constant-effect FE-DML is the best estimator when the effect is in fact constant, but its IRMSE is two to five times that of Panel-DOSE in the linear, threshold and hump designs. The boosted R-learner, which fits $\theta(\cdot)$ to the same residuals, has an IRMSE of about 0.58 in every design because the residual-on-residual regression is noisy and trees approximate smooth functions poorly with a few thousand observations. Panel-DOSE reduces the integrated error by 73–81% relative to TWFE-interaction and by 61–73% relative to the R-learner in the three heterogeneous designs, and attains the lowest IRMSE in each of them. The penalty also pays off: relative to the unpenalised sieve, GCV reduces the IRMSE by 14–40% in the heterogeneous designs. Figure 2 illustrates these differences for one sample per design.

Fourth, inference is reasonably reliable. The pointwise 95% intervals of the unpenalised sieve cover the true function at 93–94% of grid points, and those of the penalised estimator at 84–93%; the shortfall of the penalised version is largest in the threshold design, where the data-driven penalty induces smoothing bias around the steep part of the curve. Simultaneous bands cover the entire function in 70–88% of replications. All panel-DML estimators estimate the average effect with an RMSE between 0.07 and 0.11, against more than 0.5 for the linear and pooled estimators.

Table 2 shows that the ranking is robust. For the threshold design, Panel-DOSE has the lowest IRMSE for every combination of sample size and confounding strength. Its error falls from 0.35 with 50 countries to 0.15 with 200, roughly in line with root- G convergence of the sieve coefficients, whereas the pooled DML sieve and TWFE-interaction remain biased as G grows. Doubling the strength of nonlinear confounding nearly doubles the error of

Table 1: Monte Carlo results, $G = 100$, $T = 24$, $\rho = 1$.

Estimator	Effect function		Average effect		Coverage (95%)	
	IRMSE	Int. bias	Bias	RMSE	Pointwise	Uniform
<i>Design: constant $\theta(z)$</i>						
TWFE	0.631	0.631	0.631	0.640	0.00	—
TWFE-interaction	0.829	0.772	0.741	0.749	0.04	—
FE-DML (constant)	0.074	0.057	0.057	0.090	0.91	—
FE-DML + R-learner	0.572	0.051	0.059	0.097	—	—
Pooled DML-sieve	0.632	0.604	0.602	0.610	0.12	—
Panel-DOSE (unpenalised)	0.258	0.054	0.057	0.094	0.93	0.88
Panel-DOSE	0.151	0.053	0.057	0.091	0.92	0.88
<i>Design: linear $\theta(z)$</i>						
TWFE	0.936	0.572	0.525	0.538	0.20	—
TWFE-interaction	0.829	0.772	0.741	0.749	0.04	—
FE-DML (constant)	0.742	0.057	0.010	0.073	0.14	—
FE-DML + R-learner	0.587	0.054	0.056	0.100	—	—
Pooled DML-sieve	0.651	0.616	0.610	0.619	0.13	—
Panel-DOSE (unpenalised)	0.265	0.059	0.059	0.095	0.94	0.86
Panel-DOSE	0.160	0.056	0.058	0.093	0.91	0.86
<i>Design: threshold $\theta(z)$</i>						
TWFE	1.153	0.599	0.537	0.551	0.13	—
TWFE-interaction	0.870	0.656	0.708	0.718	0.16	—
FE-DML (constant)	0.986	0.069	0.007	0.079	0.07	—
FE-DML + R-learner	0.590	0.059	0.063	0.106	—	—
Pooled DML-sieve	0.681	0.630	0.623	0.632	0.19	—
Panel-DOSE (unpenalised)	0.268	0.065	0.064	0.101	0.93	0.83
Panel-DOSE	0.231	0.064	0.068	0.101	0.84	0.70
<i>Design: hump $\theta(z)$</i>						
TWFE	1.051	0.959	0.839	0.847	0.00	—
TWFE-interaction	0.829	0.772	0.741	0.749	0.04	—
FE-DML (constant)	0.494	0.239	0.119	0.139	0.30	—
FE-DML + R-learner	0.576	0.059	0.065	0.100	—	—
Pooled DML-sieve	0.701	0.652	0.615	0.623	0.12	—
Panel-DOSE (unpenalised)	0.258	0.048	0.044	0.088	0.94	0.87
Panel-DOSE	0.210	0.071	0.055	0.091	0.89	0.77

Notes: averages over 100 replications. IRMSE and integrated bias are computed over a grid of 17 points in $[0.1, 0.9]$; the lowest IRMSE in each design is in bold. Pointwise coverage is the share of grid points at which the 95% confidence interval contains $\theta(z)$; uniform coverage is the share of replications in which the 95% sup- t band contains the whole function. “—” indicates that the estimator does not provide the quantity.

TWFE-interaction but raises that of Panel-DOSE only from 0.23 to 0.29.

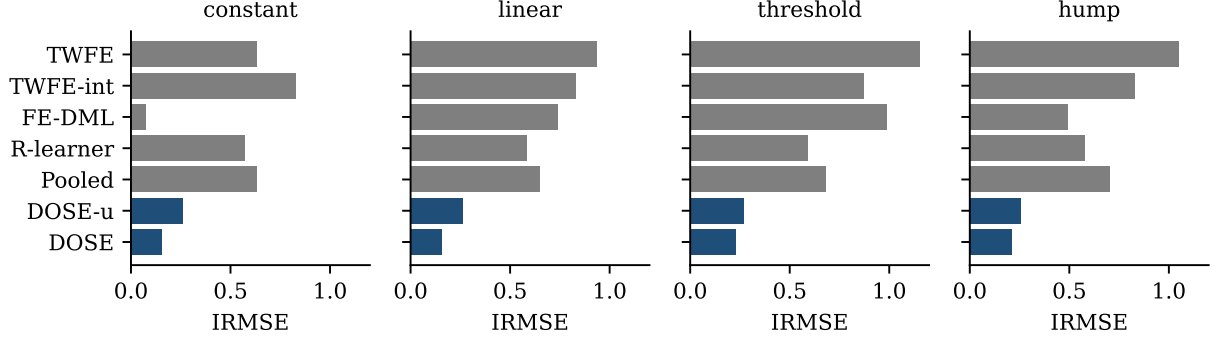


Figure 1: Integrated root-mean-squared error of the estimated effect function by design (100 replications). DOSE denotes Panel-DOSE and DOSE-u its unpenalised version.

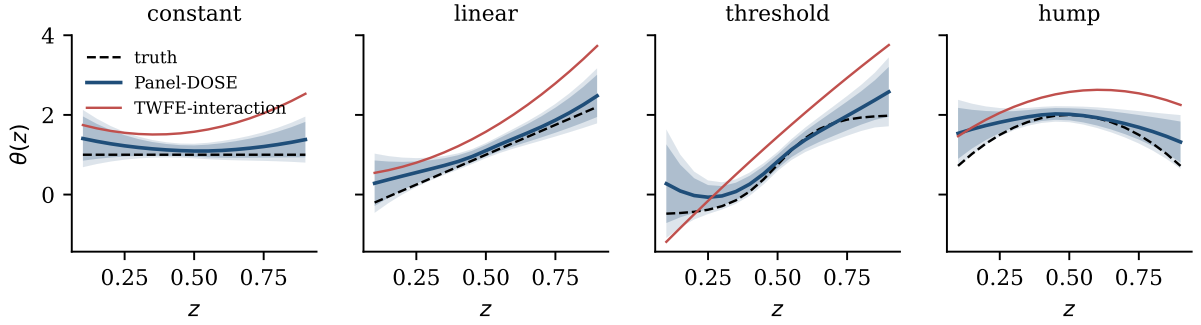


Figure 2: One simulated sample per design. Dashed black: true $\theta(z)$; blue: Panel-DOSE with 95% pointwise (dark) and simultaneous (light) bands; red: TWFE with quadratic interaction.

Table 2: Sensitivity of IRMSE to sample size and confounding strength (threshold design).

	$G = 50$	$G = 100$	$G = 100$	$G = 100$	$G = 200$
	$\rho = 1$	$\rho = 0.5$	$\rho = 1$	$\rho = 2$	$\rho = 1$
TWFE-interaction	0.889	0.559	0.870	1.657	0.867
FE-DML + R-learner	0.701	0.591	0.590	0.671	0.471
Pooled DML-sieve	0.831	0.643	0.681	0.706	0.609
Panel-DOSE	0.348	0.217	0.231	0.288	0.146

Notes: IRMSE of the estimated effect function; 50 replications per cell except $G = 100, \rho = 1$ (100 replications). ρ scales the nonlinear confounding in the outcome equation.

5. Empirical application: internet adoption and productivity growth, 1996–2025

5.1. Data and specification

The main data set combines the World Bank’s World Development Indicators (WDI, July 2026 release) with the UNDP Human Development Report 2025. The

outcome is annual growth of GDP per person employed at constant 2021 PPP prices, $Y_{it} = 100 \Delta \ln(\text{GDP}_{it}/\text{EMP}_{it})$, which WDI compiles from national accounts and the International Labour Organization’s modelled employment estimates and which is available up to 2025. The treatment is the share of the population using the internet (ITU) in year $t - 1$, measured on $[0, 1]$, so that $\theta(z)$ is the change in growth (in percentage points) associated with moving from no adoption to universal adoption; one tenth of it is the effect of a ten-percentage-point increase. Controls, all lagged one year, are log GDP per worker (conditional convergence), mean years of schooling, the gross capital formation and government consumption shares of GDP, trade openness (exports plus imports over GDP), population growth, the age-dependency ratio, the urban population share, lagged productivity growth, and mobile and fixed broadband subscriptions per capita; they follow the empirical growth literature (Barro, 1991; Mankiw et al., 1992). Mean years of schooling is available up to 2023; its 2024 value, which enters only the 2025 observations, is extrapolated from each country’s 2019–2023 trend. The sample covers 1996–2025, drops growth observations below the 1st and above the 99th percentile, and keeps countries with at least 15 usable years. The resulting unbalanced panel contains 3,806 observations on 141 countries, listed in Appendix B. Table 3 reports descriptive statistics.

To connect with the growth-accounting literature, we also use the Penn World Table (Feenstra et al., 2015), which provides capital stocks, a human-capital index and TFP but ends earlier than WDI. The main PWT panel uses version 11.0 (Groningen Growth and Development Centre, 2025) and contains 3,087 observations on 117 countries over 1996–2023. A second panel uses version 10.0, which ends in 2019, and contains 2,566 observations on 113 countries. Both use PWT labour productivity (real GDP at constant national prices per person engaged) as the outcome and add log capital per worker and the labour share to the controls. They serve as robustness checks and, because they cover the same years with different vintages of the underlying national accounts and price data, they also allow us to ask whether conclusions depend on the source and vintage of the productivity data.

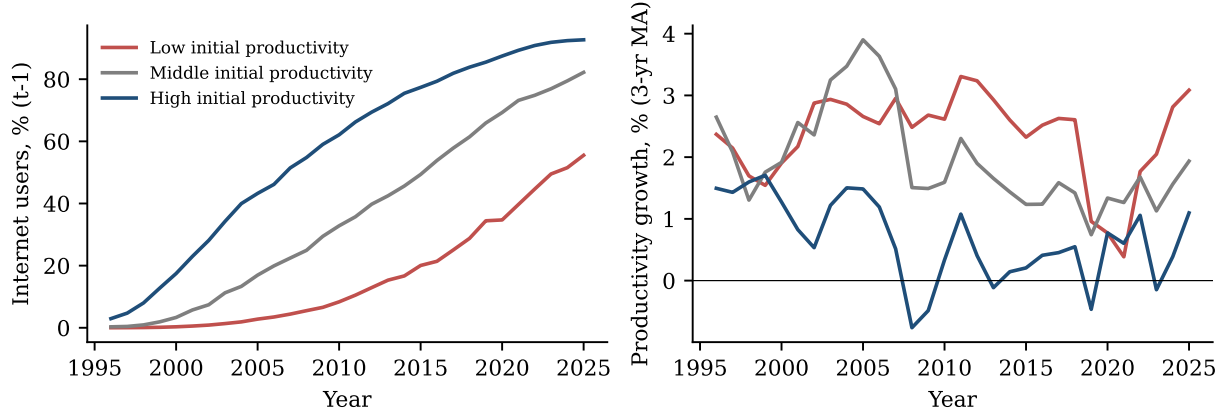


Figure 3: Internet diffusion and labour-productivity growth by tercile of initial (first-year) GDP per worker, sample averages by year, 1996–2025.

Figure 3 shows the variation that identifies the effects. Internet use diffused in all groups of countries, but at very different times: in the countries with the highest initial productivity, about half of the population was online by the mid-2000s, whereas in the least productive third the share reached 20% only in the mid-2010s and passed one half at the end of the sample. Productivity growth was highest in the low and middle groups during the 2000s, collapsed in 2009 and 2020, rebounded in 2010 and 2021, and was lowest in the high group in most years. Because common shocks such as the global financial crisis and the COVID-19 pandemic are absorbed by year effects, the estimates rely on differences across countries in the timing of adoption relative to their own productivity paths, conditional on the controls.

We estimate model (1) with two moderators that correspond to the two mechanisms discussed in Section 2: mean years of schooling (absorptive capacity) and lagged log GDP per worker (distance to the frontier). We also estimate the dose-response model (2) in internet penetration. Panel-DOSE uses LightGBM with 400 trees, five country-level folds, five repetitions of the split and cubic B-splines with four interior knots; standard errors are clustered by country. We report the average effect, the effects averaged within terciles of the moderator (low, middle, high), the high-minus-low difference, and the BLP slope.

5.2. Results

Table 4 and Figure 4 summarise the estimates.

Table 3: Descriptive statistics, 141 countries, 1996–2025.

Variable	Obs.	Mean	Std. dev.	Min	Max
Labour-productivity growth (%)	3806	1.654	3.355	-10.329	12.354
Internet users, share ($t - 1$)	3806	0.378	0.334	0.000	1.000
Mobile subscriptions per capita ($t - 1$)	3806	0.776	0.538	0.000	5.742
Fixed broadband per capita ($t - 1$)	3806	0.095	0.129	0.000	0.487
Mean years of schooling ($t - 1$)	3806	8.590	3.281	0.980	14.325
Log GDP per worker ($t - 1$)	3806	10.485	1.055	7.214	12.677
Investment share ($t - 1$)	3806	0.241	0.071	-0.024	0.611
Government consumption share ($t - 1$)	3806	0.159	0.054	0.020	0.435
Trade openness ($t - 1$)	3806	0.860	0.544	0.156	4.426
Population growth, % ($t - 1$)	3806	1.289	1.458	-10.927	21.700
Age-dependency ratio ($t - 1$)	3806	0.595	0.175	0.173	1.112
Urban population share ($t - 1$)	3806	0.607	0.214	0.109	1.000

Sources: World Development Indicators (World Bank, July 2026 release), including ITU digital-adoption series; UNDP Human Development Report 2025 (mean years of schooling).

Average effect. There is no evidence of an aggregate digital dividend in labour-productivity growth. TWFE with the same controls yields -1.23 (s.e. 0.77), constant-effect FE-DML -1.11 (1.14), and the Panel-DOSE average effect in the baseline specification is -0.94 (1.13). The last estimate implies that a ten-percentage-point increase in internet users changes subsequent annual productivity growth by -0.09 percentage points, with a 95% confidence interval of $[-0.32, 0.13]$. The estimates are therefore precise enough to rule out positive effects larger than about 0.13 percentage points per ten points of adoption, an order of magnitude smaller than the broadband effects that Czernich et al. (2011) estimate for OECD countries. The average is negative and insignificant in every WDI-based specification of Panel A, including the pre-COVID period 1996–2019 (-1.75 , s.e. 1.51) and the sample that excludes 2020–2021 (-1.01 , s.e. 1.11), so the pandemic years do not drive the result. The PWT panels agree: the average is -1.91 (s.e. 1.47) in PWT 11.0 for 1996–2023, -1.48 (1.84) in PWT 11.0 for 1996–2019 and -3.11 (2.01) in PWT 10.0, all insignificant, and restricting the WDI data to the countries and years of the PWT 11.0 panel gives -1.44 (1.72). The absence of an average dividend is therefore robust to the source and vintage of the data.

Schooling. With mean years of schooling as the moderator (Panel A of Table 4 and Figure 4a), the estimated function is flat and close to zero: it lies between -2.3 and 1.1 over the 5th to 95th percentiles of schooling (2.6 to 13.0 years), the high-minus-low difference is 1.15 (s.e. 2.74), the BLP slope is 0.13 (s.e. 0.38) per year of schooling, and the simultaneous band contains a constant function. The absence of heterogeneity holds across learners, sieves, lags, samples and data sources (Figure 5). The only specification in which the difference reaches the 10% level is the one without Mundlak features, which the simulations show to be biased.

Distance to the frontier. With lagged log GDP per worker as the moderator (Panel B and Figure 4b), the evidence is more intriguing but fragile. In the baseline specification the average effect is positive in the least productive third of country-years (7.42, s.e. 2.97) and negative in the most productive third (-2.07 , s.e. 1.87), and the difference of -9.49 (s.e. 3.46) is significant at the 1% level. This is the pattern predicted by “leapfrogging” arguments, in which the internet lowers information and transaction costs most where they are initially highest. Three observations, however, counsel against taking it at face value. First, the estimated function is not monotone: it oscillates between about -4.5 and 4.4 below 25,000 dollars per worker, the BLP slope (-2.20 , s.e. 1.39) is not significant, and the 95% simultaneous band contains a constant function. Second, the tercile difference keeps its negative sign in all WDI-based robustness checks but shrinks to between -3.0 and -7.4 and is not significant in any of them. Third, and most tellingly, the result depends on the productivity data. Restricting the WDI data to the countries and years of the PWT 11.0 panel gives a similar difference (-8.77 , s.e. 4.26), so the sample does not explain it. With PWT 11.0 itself, however, the difference is small, positive and insignificant, both for 1996–2023 (2.55, s.e. 3.37) and for 1996–2019 (3.11, s.e. 4.14), and it is close to zero for TFP growth (1.29, s.e. 3.61). With PWT 10.0 for 1996–2019 the pattern is reversed and strong: the effect is -17.19 (s.e. 5.54) in the least productive third and close to zero in the most productive third, a difference of $+17.50$ (s.e. 5.84) that is significant at the 1% level. The comparison of the two PWT vintages over identical years is the cleanest test of fragility: a revision of

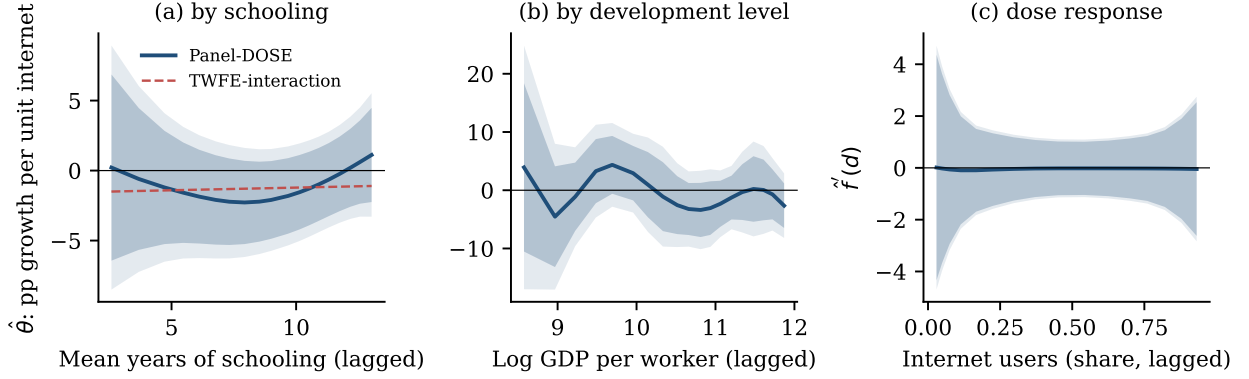


Figure 4: Estimated effect of lagged internet adoption on labour-productivity growth, WDI panel 1996–2025. (a) $\hat{\theta}(z)$ by mean years of schooling, with the TWFE linear-interaction estimate (dashed); (b) $\hat{\theta}(z)$ by lagged log GDP per worker; (c) marginal effect $\hat{f}'(d)$ in the dose-response model. Dark bands: 95% pointwise; light bands: 95% simultaneous. Grids span the 5th to 95th percentiles of the moderator (20th to 95th in panel c).

the underlying national accounts and price data between the two releases is enough to turn a highly significant J-curve pattern into a null result.

The linear benchmark illustrates why this matters. A TWFE regression with a linear interaction between internet use and log GDP per worker produces an interaction coefficient of -1.91 with a standard error of 0.52 ($t \approx 3.7$), that is, apparently strong evidence that the returns to adoption fall with development. The debiased estimates show that this precision is largely an artefact of the linear control function: once the controls enter flexibly, the gradient is imprecisely estimated, it is not robust to reasonable changes in the specification, and its sign and significance depend on the source and vintage of the data.

Dose response. The dose-response model (Figure 4c) asks whether the returns to adoption change with the level of adoption itself. The penalty selected by GCV shrinks f to an almost linear function, and the estimated marginal effect $\hat{f}'(d)$ is within 0.1 of zero over the whole range from the 20th to the 95th percentile of internet penetration; the 95% simultaneous band, whose half-width is about one percentage point, contains the zero function. There is thus no indication of either increasing returns (network effects) or diminishing returns (saturation) to internet adoption in aggregate productivity growth.

Table 4: Heterogeneous effects of internet adoption on labour-productivity growth.

Specification	Average effect		Tercile of moderator			High –	BLP
	Est.	(s.e.)	Low	Middle	High	low	slope
<i>Panel A. Moderator: mean years of schooling</i>							
Baseline	−0.94	(1.13)	−1.02 (2.18)	−1.93 (1.35)	0.13 (1.37)	1.15 (2.74)	0.13 (0.38)
Random forest nuisance	−1.27	(1.13)	−3.11 (2.00)	−1.15 (1.12)	0.46 (1.53)	3.56 (2.73)	0.47 (0.37)
Two-year lag of internet	−1.42	(1.21)	−3.28 (2.27)	−1.48 (1.23)	0.49 (1.35)	3.77 (2.81)	0.53 (0.42)
Without lagged growth	−0.70	(1.16)	−0.35 (2.29)	−1.90 (1.41)	0.16 (1.38)	0.50 (2.77)	0.20 (0.40)
Three interior knots	−0.96	(1.12)	−0.91 (2.21)	−2.05 (1.33)	0.08 (1.37)	0.98 (2.76)	0.13 (0.38)
Six interior knots	−0.98	(1.12)	−1.11 (2.20)	−2.00 (1.35)	0.16 (1.38)	1.27 (2.79)	0.13 (0.38)
Unpenalised sieve	−1.34	(1.24)	−1.19 (2.54)	−2.81 (1.77)	−0.03 (1.58)	1.16 (3.06)	0.13 (0.38)
Broadband as treatment	−4.51	(3.55)	−6.63 (7.40)	−4.49 (3.46)	−2.42 (2.19)	4.21 (8.22)	0.63 (1.42)
No Mundlak means	−0.43	(0.96)	−2.55 (1.76)	−0.29 (0.95)	1.57 (1.25)	4.12* (2.34)	0.60* (0.34)
Pre-COVID sample, 1996-2019	−1.75	(1.51)	−2.42 (2.73)	−1.75 (1.50)	−1.07 (1.55)	1.35 (3.22)	0.19 (0.48)
Excluding 2020-2021	−1.01	(1.11)	−0.70 (2.06)	−2.37* (1.40)	0.03 (1.40)	0.73 (2.65)	0.21 (0.39)
WDI data, PWT 11.0 countries, 1996-2023	−1.44	(1.72)	−2.19 (4.31)	−1.75 (1.96)	−0.37 (1.83)	1.82 (5.31)	0.47 (0.60)
PWT 11.0 sample, 1996-2023	−1.91	(1.47)	−5.75 (3.84)	−0.61 (1.38)	0.62 (1.10)	6.37 (4.18)	3.92 (2.66)
PWT 11.0, 1996-2019	−1.48	(1.84)	−4.40 (4.66)	−0.45 (1.66)	0.42 (1.14)	4.82 (5.08)	2.95 (3.14)
PWT 10.0 sample, 1996-2019	−3.11	(2.01)	−6.98 (5.40)	−1.48 (1.73)	−0.86 (0.98)	6.12 (5.65)	2.95 (3.19)
PWT 11.0: outcome TFP growth	−2.41	(1.59)	−4.52 (3.20)	−2.03 (1.57)	−0.66 (1.39)	3.86 (3.69)	2.95 (2.59)
<i>Panel B. Moderator: lagged log GDP per worker</i>							
Baseline	1.06	(1.36)	7.42** (2.97)	−2.18 (1.88)	−2.07 (1.87)	−9.49*** (3.46)	−2.20 (1.39)
Random forest nuisance	−1.00	(1.17)	0.10 (2.40)	−0.15 (1.43)	−2.94** (1.44)	−3.04 (2.94)	−1.48 (1.30)
Two-year lag of internet	0.45	(1.96)	4.90 (5.31)	−1.06 (1.73)	−2.51 (1.81)	−7.41 (5.58)	−1.38 (1.59)
Without lagged growth	0.38	(1.72)	5.06 (4.59)	−1.99 (1.78)	−1.93 (1.59)	−6.99 (4.91)	−1.94 (1.55)
Pre-COVID sample, 1996-2019	−0.07	(2.08)	3.97 (4.84)	−2.42 (2.64)	−1.77 (2.13)	−5.74 (5.21)	−2.19 (1.61)
Excluding 2020-2021	−0.50	(1.40)	2.91 (3.40)	−2.34 (2.00)	−2.07 (1.88)	−4.98 (3.94)	−2.49* (1.39)
WDI data, PWT 11.0 countries, 1996-2023	0.80	(1.47)	6.26* (3.44)	−1.34 (1.82)	−2.51 (2.71)	−8.77** (4.26)	−1.56 (2.30)
PWT 11.0 sample, 1996-2023	−1.40	(1.23)	−3.14 (2.48)	−0.46 (1.26)	−0.59 (1.80)	2.55 (3.37)	1.39 (1.54)
PWT 11.0, 1996-2019	−1.39	(1.49)	−3.17 (2.97)	−0.96 (1.48)	−0.06 (2.13)	3.11 (4.14)	1.62 (1.93)
PWT 10.0 sample, 1996-2019	−6.28***	(2.20)	−17.19*** (5.54)	−1.93 (1.79)	0.30 (2.14)	17.50*** (5.84)	4.22*** (2.07)
PWT 11.0: outcome TFP growth	−2.05	(1.40)	−2.95 (2.99)	−1.56 (1.60)	−1.65 (1.94)	1.29 (3.61)	−0.02 (1.76)

Notes: Panel-DOSE estimates. The dependent variable is annual labour-productivity growth (%) unless stated otherwise, and the treatment is the lagged share of internet users (0–1). The baseline sample is the WDI panel (141 countries, 1996–2025). “WDI data, PWT 11.0 countries” restricts it to the countries of the PWT 11.0 panel and to 1996–2023. The PWT rows use the Penn World Table 11.0 panel (117 countries, 1996–2023, or its 1996–2019 subsample) or the 10.0 panel (113 countries, 1996–2019); in them the moderators are the PWT human-capital index (Panel A) and log PWT labour productivity (Panel B), and the controls add log capital per worker and the labour share. Terciles are defined over country-year observations. Standard errors (in parentheses) are clustered by country. *, **, *** denote significance at 10%, 5% and 1%.

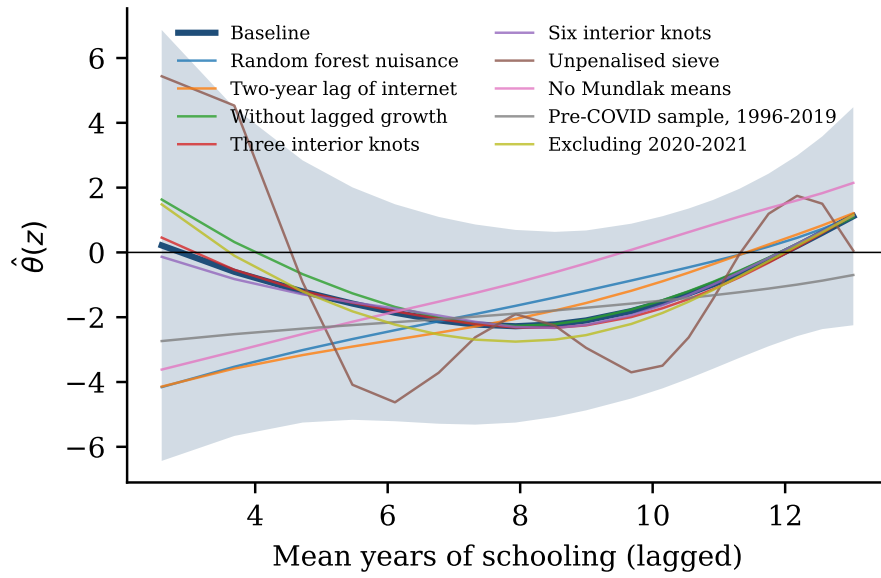


Figure 5: Robustness of $\hat{\theta}(z)$ by mean years of schooling (WDI panel) to the learner, the sieve, the timing of adoption, the control set, the sample period and the treatment of fixed effects. The shaded band is the 95% pointwise band of the baseline.

6. Discussion

6.1. Interpretation

Three features of the empirical results deserve emphasis. First, over three decades and 141 economies, the diffusion of the internet has not been followed by faster measured labour-productivity growth. The point estimates are small and negative, and they are precise enough to exclude effects of the size often invoked in policy discussions. This is consistent with the aggregate productivity paradox and with the J-curve argument of Brynjolfsson et al. (2021): if the gains require complementary intangible investment that is poorly measured, the measured productivity dividend of a general-purpose technology may be small for a long time. It is also consistent with Niebel (2018) and with the heterogeneous findings surveyed by Vu et al. (2020). Access, as measured by the share of internet users, is a coarse proxy for the intensity and quality of use, and the gains may accrue through channels—consumer surplus, variety, time savings—that productivity statistics do not capture (Byrne et al., 2016; Syverson, 2017).

Second, formal schooling does not condition the effect. The theory of absorptive capacity (Nelson and Phelps, 1966; Benhabib and Spiegel, 1994; Cohen and Levinthal, 1990) predicts larger returns in better-educated economies, but the estimated function is flat across the whole distribution of schooling and the data reject even moderate gradients. Years of schooling may capture the skills that are complementary to digital technologies only imperfectly (Sianesi and Van Reenen, 2003), and complementarities may operate through managerial and organisational capabilities that aggregate data do not measure.

Third, heterogeneity along the development dimension is not robustly identified. Depending on the productivity data, the estimates suggest leapfrogging (larger effects in low-productivity economies, WDI), a J-curve in which the least productive economies experience a transitional drag (PWT 10.0), or no gradient at all (PWT 11.0). Neither pattern survives the full set of checks, and the linear TWFE interaction that seems to settle the question owes its precision to a restrictive control function. We view this as the main substantive

lesson of the application: conclusions about where the digital dividend arrives are sensitive to data vintages and specification choices, and flexible estimators with honest uncertainty quantification are needed to reveal that sensitivity.

6.2. Policy implications

The results caution against treating connectivity targets as productivity policy. Across the income distribution, expanding internet access has not been followed by faster measured productivity growth within the horizon we observe. This does not imply that such investments are undesirable, since connectivity has consumer-surplus, inclusion and service-delivery benefits that productivity statistics do not capture (World Bank, 2016; Goldfarb and Tucker, 2019). It does imply that a productivity dividend, if it exists, is contingent on complementary investment—in firm capabilities, management practices, skills and the reallocation of resources towards digitally intensive activities—and that policies promoting access should be evaluated on their own merits rather than justified by expected aggregate productivity gains. The fragility of the heterogeneity estimates also suggests that governments should be wary of cross-country benchmarks that claim to identify the stage of development at which digital investment pays off.

6.3. Limitations

The analysis has limitations that also point to extensions. The identifying assumptions are those of static fixed-effects models augmented by the correlated-random-effects restriction in the nuisance step. Strict exogeneity rules out feedback from productivity shocks to future adoption; lagging the treatment by one or two years mitigates reverse causality but does not eliminate it, and the inclusion of lagged productivity growth is subject to a Nickell-type bias that is small with $T \approx 27$ but not zero (Nickell, 1981). Results are, however, similar when lagged growth is excluded. An instrumental-variable version of Panel-DOSE, in which the orthogonalised treatment is replaced by an orthogonalised instrument in the spirit of Czernich et al. (2011), is a natural extension. Second, the most recent observations rest on provisional

national accounts and on the ILO’s modelled employment estimates, and the 2024 value of schooling is extrapolated. The similarity of the full-sample and pre-COVID estimates suggests that these data issues do not drive the average results, but the comparison of PWT vintages shows that data revisions can change conclusions about heterogeneity. Third, internet penetration measures access rather than the intensity or quality of use; measures of firm-level digital adoption or data-intensive investment would sharpen the interpretation. Fourth, the asymptotic theory treats the sieve dimension as fixed, so the confidence bands for $\theta(z)$ are centred at the best spline approximation; the simulation evidence indicates that the smoothing bias induced by the data-driven penalty can reduce pointwise coverage for sharply nonlinear functions, and the unpenalised version or low-dimensional functionals (group averages, BLP slopes) should be preferred when coverage is paramount. Finally, as with any aggregate analysis, the estimates describe country-level associations and need not carry over to firms or regions.

7. Conclusion

This paper has proposed Panel-DOSE, an estimator for the function that links the effect of a continuous digital treatment to the characteristics of the adopting economy in fixed-effects panels. The estimator combines Mundlak-augmented machine-learning nuisance functions, cross-fitting by country, a two-way within transformation of the orthogonalised residuals and a penalised spline sieve. Its estimating equation is Neyman-orthogonal and exactly invariant to additive country and year errors in the nuisance fits, and when the moderator is a control variable the estimator needs only two nuisance functions whatever the flexibility of the sieve. The procedure provides the effect function with pointwise and simultaneous bands, group-average effects and a best-linear-projection summary, and it is simple to implement with off-the-shelf learners.

The Monte Carlo evidence shows that these ingredients matter. Linear fixed-effects models are badly biased when the control function is nonlinear, pooled machine-learning

estimators that ignore the panel structure inherit the bias of the fixed effects, and a boosted R-learner applied to the same residuals is noisier than the sieve. Panel-DOSE attains the lowest integrated error in every design with a heterogeneous effect.

Applied to 141 economies over 1996–2025, the estimator finds no aggregate digital dividend in labour-productivity growth: the average effect of internet adoption is small, negative and insignificant, and it does not vary with schooling. Heterogeneity along the development dimension appears in some specifications but changes sign with the source of the productivity data and disappears in the latest vintage of the Penn World Table, whereas a linear fixed-effects interaction reports it with spurious precision. The digital dividend, in other words, does not arrive automatically with connectivity, and claims about where it arrives should be treated with caution. Future work could extend the estimator to instrumental variables and dynamic panels, apply it to firm- and region-level data on the intensity of digital use, and examine the moderating roles of institutions, competition and data governance, all of which are central to the digital economy.

Data and code availability

The data are publicly available from the World Bank’s World Development Indicators, the UNDP Human Development Report 2025 and the Penn World Table (versions 11.0 and 10.0). The Python code that implements Panel-DOSE, downloads the data and reproduces all tables and figures is included in the replication package accompanying this article.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRedit authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review & editing.

Appendix A. Proofs

Throughout, the year effect is a function of the time index t , which is an element of F_{it} , so λ_t is absorbed by ℓ and $\mathbb{E}[\lambda_t \mid F_{it}] = \lambda_t$. The two-way operator $\mathcal{M}$ is the orthogonal projection onto the complement of the space spanned by country and year indicators; it is symmetric and idempotent, and $\mathcal{M}c = 0$ for any vector of the form $c_{it} = c_i + c_t$. For country i , collect the T_i observations into Y_i , Φ_i ($T_i \times K$) and F_i , and write $V_i = \Phi_i - m_0(F_i)$ and $e_i = Y_i - \ell_0(F_i) - V_i\beta_0$. Under Assumptions 1–2 and $r_K = 0$, $e_{it} = a_i + \varepsilon_{it}$ with $a_i = \alpha_i - h(\bar{X}_i, \bar{D}_i)$. In the asymptotic arguments below, the within transformation is treated as country demeaning $Q_i = I_{T_i} - T_i^{-1}\iota\iota'$; the additional year demeaning is asymptotically negligible because year means converge to their population counterparts at rate $G^{-1/2}$ and λ_t is already absorbed by ℓ .

Proof of Proposition 1. (i) For directions $\delta_\ell(F)$ and $\delta_m(F)$,

$$\psi(\beta_0; \ell_0 + r\delta_\ell, m_0 + r\delta_m) = \mathbb{E}[(V_i - r\delta_{m,i})' Q_i (e_i - r\delta_{\ell,i} + r\delta_{m,i}\beta_0)].$$

Differentiating at $r = 0$ gives $-\mathbb{E}[\delta'_{m,i} Q_i e_i] + \mathbb{E}[V_i' Q_i (\delta_{m,i}\beta_0 - \delta_{\ell,i})]$. In the first term, $Q_i e_i = Q_i \varepsilon_i$ because a_i is constant within country, and $\mathbb{E}[\varepsilon_{it} \mid F_{i1}, \dots, F_{iT_i}] = 0$ by Assumption 1, since F_{is} is a function of (D_i, X_i) ; hence the term is zero. In the second term, $Q_i w_i$ is a

function of $(F_{i1}, \dots, F_{iT_i})$ for $w_i = \delta_{m,i}\beta_0 - \delta_{\ell,i}$, and $\mathbb{E}[V_{it} \mid F_{i1}, \dots, F_{iT_i}] = 0$ by Assumption 3; hence it is also zero.

(ii) $\mathcal{M}(Y - \ell - c) = \mathcal{M}(Y - \ell) - \mathcal{M}c = \mathcal{M}(Y - \ell)$ and $\mathcal{M}(\Phi - m - d) = \mathcal{M}(\Phi - m)$ whenever c and each column of d are sums of country and year constants, because $\mathcal{M}$ annihilates the span of the country and year indicators. The sample and population moments are therefore unchanged for every β . $\square$

Proof of Proposition 2. If Z_{it} is a coordinate of F_{it} , then $b_k(Z_{it})$ is F_{it} -measurable and $\mathbb{E}[D_{it}b_k(Z_{it}) \mid F_{it}] = b_k(Z_{it})\mathbb{E}[D_{it} \mid F_{it}]$. Consequently $\Phi_{it} - m(F_{it}) = b(Z_{it})\{D_{it} - \mu(F_{it})\}$. In the cross-fitted version, replacing μ by $\hat{\mu}^{(-l)}$ gives the residuals in Algorithm 1. $\square$

Proof of Proposition 3. Write $\tilde{\Phi} = \mathcal{M}(\Phi - \hat{m})$, $\tilde{Y} = \mathcal{M}(Y - \hat{\ell})$, and let $\tilde{\Phi}^0, \tilde{Y}^0$ be the same objects evaluated at the true nuisance functions. From (4),

$$\sqrt{G}(\hat{\beta}_\lambda - \beta_K) = \left(\frac{1}{G}\tilde{\Phi}'\tilde{\Phi} + \lambda \frac{n}{G}P \right)^{-1} \left\{ \frac{1}{\sqrt{G}}\tilde{\Phi}'(\tilde{Y} - \tilde{\Phi}\beta_K) - \sqrt{G}\lambda \frac{n}{G}P\beta_K \right\}.$$

The last term vanishes because $n/G \leq \bar{T}$ and $\lambda_G\sqrt{G} \rightarrow 0$. Let $\Delta_m = \hat{m} - m_0$ and $\Delta_\ell = \hat{\ell} - \ell_0$ on the evaluation fold. Then

$$\frac{1}{\sqrt{G}}\tilde{\Phi}'(\tilde{Y} - \tilde{\Phi}\beta_K) = \frac{1}{\sqrt{G}} \sum_i V_i' Q_i e_i + R_1 + R_2 + R_3,$$

where

$$\begin{aligned} R_1 &= -G^{-1/2} \sum_i \Delta'_{m,i} Q_i e_i, & R_2 &= G^{-1/2} \sum_i V_i' Q_i (\Delta_{m,i} \beta_K - \Delta_{\ell,i}), \\ R_3 &= -G^{-1/2} \sum_i \Delta'_{m,i} Q_i (\Delta_{m,i} \beta_K - \Delta_{\ell,i}). \end{aligned}$$

Because the nuisance functions are estimated on countries outside the evaluation fold, conditionally on the training sample the summands of R_1 and R_2 are independent across countries with mean zero (by the arguments in the proof of Proposition 1), and their conditional vari-

ances are bounded by a constant times $\|\Delta_m\|_2^2$ and $\|\Delta_m\|_2^2 + \|\Delta_\ell\|_2^2$; hence $R_1, R_2 = o_p(1)$ by Chebyshev’s inequality under Assumption 4(b) and the consistency of both nuisance estimators. By the Cauchy–Schwarz inequality and $\|Q_i\| \leq 1$, $\|R_3\| \leq C\sqrt{G} \|\Delta_m\|_2(\|\Delta_m\|_2 + \|\Delta_\ell\|_2)$ for a constant C depending on $\bar{T}$, β_K and the bounded basis, which is $o_p(1)$ under Assumption 4(b); by Proposition 2, $\|\Delta_m\|_2$ is bounded by a constant times $\|\hat{\mu} - \mu\|_2$. By the same arguments, $G^{-1}\tilde{\Phi}'\tilde{\Phi} = G^{-1}\sum_i V_i'Q_iV_i + o_p(1) \xrightarrow{p} J$. The Lindeberg–Feller central limit theorem for independent clusters with bounded fourth moments gives $G^{-1/2}\sum_i V_i'Q_ie_i \xrightarrow{d} N(0, \Omega)$, since $V_i'Q_ie_i = V_i'Q_i\varepsilon_i = s_i$ and $\mathbb{E}[s_i] = 0$. Slutsky’s theorem yields the stated limit. Consistency of $G\hat{V}$ follows by standard arguments for cluster-robust variance estimators (Cameron and Miller, 2015), because $\hat{u}_{it} - u_{it}$ and $\tilde{\Phi}_{it} - \tilde{\Phi}_{it}^0$ vanish in mean square. Finally, for a fixed number S of sample splits, each $\hat{\beta}_s$ satisfies the same linear representation with the same leading term, so the median $\hat{\beta}$ does too, and the variance adjustment is asymptotically negligible (Chernozhukov et al., 2018). $\square$

Appendix B. Data and implementation details

Variables. Main panel (1996–2025). From the World Development Indicators (July 2026 release, retrieved through the World Bank API): GDP per person employed at constant 2021 PPP dollars (SL.GDP.PCAP.EM.KD); individuals using the internet (IT.NET.USER.ZS); mobile cellular (IT.CEL.SETS.P2) and fixed broadband (IT.NET.BBND.P2) subscriptions per 100 people; gross capital formation (NE.GDI.TOTL.ZS), government final consumption (NE.CON.GOVT.ZS), exports (NE.EXP.GNFS.ZS) and imports (NE.IMP.GNFS.ZS) as percentages of GDP; population growth (SP.POP.GROW); the urban population share (SP.URB.TOTL.IN.ZS); and the age-dependency ratio (SP.POP.DPND). Mean years of schooling is taken from the UNDP Human Development Report 2025 composite-indices time series (1990–2023); the 2024 value is the 2023 value plus the country’s 2019–2023 linear trend. Percentages are divided by 100. Broadband subscriptions are set to zero in the years before a country’s first reported value. Regional and income aggregates are excluded.

PWT panels. From the Penn World Table 11.0 (1996–2023) and 10.0 (1996–2019), with identical definitions: labour productivity is real GDP at constant national prices (`rgdpna`) divided by persons engaged (`emp`); capital per worker uses the capital stock at constant national prices (`rnna`); TFP growth is the log change of `rtfpna`; the human capital index is `hc`; the investment and government shares are `csh_i` and `csh_g`; openness is `csh_x-csh_m`; the labour share is `labsh`. The digital-adoption series are the ITU data distributed by Our World in Data. All controls and the treatment enter with a one-year lag in both panels.

Countries. Table B.5 lists the 141 countries of the main panel by tercile of initial GDP per worker, the grouping used in Figure 3. (The terciles in Table 4 are defined over country-year observations of lagged productivity, so countries can move between groups over time.)

Table B.5: Countries in the main estimation sample, by tercile of initial GDP per worker.

Group	Countries
Low	Armenia; Azerbaijan; Bangladesh; Benin; Bolivia; Burkina Faso; Cambodia; Cameroon; Central African Republic; Chad; China; Comoros; Congo, Dem. Rep.; Gambia, The; Georgia; Ghana; Guinea; Guinea-Bissau; Haiti; Honduras; India; Indonesia; Kenya; Kyrgyz Republic; Lao PDR; Lesotho; Madagascar; Mali; Moldova; Mongolia; Morocco; Mozambique; Nepal; Nicaragua; Niger; Pakistan; Philippines; Rwanda; Senegal; Sierra Leone; Solomon Islands; Sri Lanka; Tajikistan; Tanzania; Togo; Tonga; Uganda; Uzbekistan; Viet Nam; Zambia; Zimbabwe
Middle	Albania; Angola; Belarus; Belize; Bosnia and Herzegovina; Botswana; Brazil; Bulgaria; Cabo Verde; Chile; Colombia; Congo, Rep.; Costa Rica; Croatia; Dominican Republic; Ecuador; Egypt, Arab Rep.; El Salvador; Estonia; Fiji; Guatemala; Iran, Islamic Rep.; Kazakhstan; Korea, Rep.; Latvia; Lithuania; Malaysia; Mauritania; Mauritius; Namibia; North Macedonia; Panama; Paraguay; Peru; Poland; Romania; Russian Federation; Sao Tome and Principe; Serbia; Slovak Republic; South Africa; Thailand; Tunisia; Turkiye; Ukraine; Uruguay
High	Algeria; Argentina; Australia; Austria; Bahamas, The; Bahrain; Belgium; Brunei Darussalam; Canada; Cyprus; Czechia; Denmark; Finland; France; Gabon; Germany; Greece; Hong Kong SAR, China; Hungary; Iceland; Ireland; Israel; Italy; Japan; Kuwait; Luxembourg; Malta; Mexico; Montenegro; Netherlands; New Zealand; Norway; Oman; Portugal; Qatar; Saudi Arabia; Singapore; Slovenia; Spain; Sweden; Switzerland; United Arab Emirates; United Kingdom; United States

Implementation. The nuisance learners are LightGBM gradient-boosted trees (Ke et al., 2017) with 15 leaves, a minimum of 20 observations per leaf (10 in the simulations), row and column subsampling of 80% and a learning rate of 0.03; the random-forest robustness check uses 400 trees (Breiman, 2001) with a minimum leaf size of 5 and half of the features

considered at each split. The feature set F_{it} contains the lagged controls, their country means, the country means of the treatment and the moderator, and the calendar year. The B-spline knots are placed at equally spaced quantiles of the moderator between its 0.5th and 99.5th percentiles, and values outside this range are assigned to the boundary. The GCV grid contains zero and 36 values of λ equally spaced on the log scale between 10^{-4} and 10^3 . Simultaneous bands use 20,000 Gaussian draws. With two nuisance functions, five folds and five repetitions, a full Panel-DOSE fit on the empirical sample takes about 30 seconds on a single CPU core; by Proposition 2, re-estimating the final stage with a different sieve takes a fraction of a second. The Python implementation, the data and the scripts that reproduce every table and figure are part of the replication package.

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